

SRAI

PU-B01-C04

Matrix Algebra and Linear Systems

Educational Exercises and Complete Solutions

Version	v1.0
Control status	Production candidate
Purpose	Practice, assessment and instructor reference

Controlled learning asset. Advancement to public release requires completion of the remaining SRAI publication gates.

How to use this asset

The exercises progress from matrix fluency to numerical reasoning and professional interpretation. Learners should state dimensions, show intermediate work, verify results and distinguish solving equations from validating a real-world model.

Level	Focus	Exercises
A	Operations and interpretation	1-4
B	Systems, residuals and rank	5-8
C	Conditioning, least squares and decisions	9-12

Part I - Exercises

1. Dimensions and meaning

Let A be 3×4 and B be 4×2 . State the dimensions of A^T , AB and BA . Identify any undefined product and explain why.

2. Addition and transpose

For $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 1 \\ -1 & 2 \end{bmatrix}$, compute $A+B$, A^T and $(A+B)^T$. Verify $(A+B)^T = A^T + B^T$.

3. Ordered multiplication

Using $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, compute AB and BA . Use the result to explain why matrix multiplication is not generally commutative.

4. Trace and determinant

For $C = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, compute $\text{tr}(C)$ and $\det(C)$. Decide whether C is singular and justify your answer.

5. Solve by elimination

Solve $x+y+z=6$, $2x-y+z=3$ and $x+2y-z=2$ by Gaussian elimination. Verify the solution by substitution.

6. Why pivoting matters

For $0.0001x+y=1$ and $x+y=2$, perform the first elimination step without row swapping, then explain why partial pivoting swaps the equations.

7. Signed residual

Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, $b = [9, 8]^T$ and $\hat{x} = [2, 3]^T$. Compute $r = b - A\hat{x}$. Interpret the result without claiming more than the residual establishes.

8. Rank and identifiability

Compare $A_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ with $A_2 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Determine their ranks and explain the implication for uniquely solving $Ax=b$.

9. Conditioning and sensitivity

Solve $x+y=2$ and $x+1.0001y=2.0001$. Then change the second right-hand side to 2.0002 and resolve. Explain the observed sensitivity.

10. Least squares

The equations $x=1$, $x=2$ and $x=2$ cannot all hold exactly. Find the scalar x minimizing the sum of squared residuals and state the residual vector using $b-Ax$.

11. Resource-allocation system

Two programmes require 100 staff-days and 80 equipment-days. Programme A uses (2,1) units per package and Programme B uses (1,2). Form and solve the allocation system. Check non-negativity, residual and interpretation.

12. AI reasoning clinic

Evaluate: 'The residual is zero, therefore the policy model is accurate.' Rewrite the statement as a defensible professional conclusion and list four additional validation checks.

Part II - Complete solutions

Solution 1

A^T is 4×3 . AB is 3×2 because the inner dimensions 4 and 4 agree. BA is undefined because B has 2 columns while A has 3 rows.

Solution 2

$A+B = \begin{bmatrix} 6 & -1 \\ 2 & 6 \end{bmatrix}$, $A^T = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$, and $(A+B)^T = \begin{bmatrix} 6 & 2 \\ -1 & 6 \end{bmatrix}$. Direct addition of A^T and B^T gives the same matrix.

Solution 3

$AB = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix}$ while $BA = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$. The products differ because multiplication composes row-column transformations in an ordered sequence.

Solution 4

$\text{tr}(C) = 4+3=7$. $\det(C) = 4 \cdot 3 - 2 \cdot 1 = 10$. Since the determinant is nonzero, C is nonsingular.

Solution 5

Elimination gives $y+z=3$ and $3y-2z=-1$. Hence $z=2$, $y=1$ and $x=3$. Substitution returns 6, 3 and 2 respectively.

Solution 6

Without swapping, the multiplier is 10,000, which magnifies rounding. Partial pivoting uses the second equation first because $|1| > |0.0001|$, producing a safer pivot.

Solution 7

$A\hat{x} = [9, 8]^T$, so $r = [0, 0]^T$. The proposed solution satisfies the stated equations numerically; this does not establish that the equations model reality correctly.

Solution 8

A_1 has rank 2 because $\det(A_1) = -2$. A_2 has rank 1 because its second row is twice its first. A full-rank square system may have a unique solution; the rank-deficient system cannot identify two independent unknowns uniquely.

Solution 9

The original solution is (1,1). After changing the second right-hand side, subtracting the first equation gives $0.0001y = 0.0002$, so $y=2$ and $x=0$. A change of 0.0001 in data causes an order-one change in the solution: the system is ill-conditioned.

Solution 10

With $A = [1, 1, 1]^T$ and $b = [1, 2, 2]^T$, the least-squares estimate is the mean $x = 5/3$. The signed residual is $[-2/3, 1/3, 1/3]^T$, whose entries sum to zero.

Solution 11

The system is $2x+y=100$ and $x+2y=80$. Solving gives $x=40$, $y=20$. Both allocations are nonnegative and the residual is zero. Operational adoption still requires valid coefficients, units, capacity constraints and stakeholder validation.

Solution 12

Defensible conclusion: 'A zero residual shows that the computed solution satisfies the specified equations at the reported precision.' Additional checks include data quality and units, model specification, rank and conditioning, sensitivity to perturbations, feasibility constraints and domain review.

Assessment rubric

- 4 - correct result, transparent reasoning and appropriate interpretation.
- 3 - correct method with a minor computational or communication error.
- 2 - partial mathematical structure but incomplete execution.
- 1 - relevant attempt showing limited progress.
- 0 - no relevant work or unsupported answer.